

機器可檢驗通用工程科學 (MC-UES)

Machine-Checkable Universal Engineering Science (MC-UES)

文件地位：唯一 **canonical specification** 文件性質：可機械檢查、可重播、可審計、可版本治理之正式規格

0. 文檔前言與語義鎖定

0. Document Preface and Semantic Lock

0.1 文件目的

中文：本文件定義機器可檢驗通用工程科學之正式規格。其目標不是提供一般敘述，而是建立一套可被機器解析、判定、驗證、重播與審計的工程語義系統。 English: This document defines the formal specification of Machine-Checkable Universal Engineering Science. Its goal is not to provide general narration, but to establish an engineering semantic system that can be machine-parsed, judged, verified, replayed, and audited.

0.2 文件地位

中文：本文件為唯一語義來源。任何外部摘要、衍生說明、二次轉述、模型內部推演或口語化解釋，皆不得高於本文件之正式定義。 English: This document is the sole semantic source. Any external summary, derivative explanation, secondary retelling, internal model inference, or informal interpretation must not take precedence over the formal definitions in this document.

0.3 語義鎖定原則

中文：本文件採用語義鎖定原則。中文文本為主語義層，英文文本為對齊別名層。英文不是另一套概念系統，僅作為同一規格的對應映射。 English: This document adopts the semantic locking principle. The Chinese text is the primary semantic layer, and the English text is the aligned alias layer. English is not a separate conceptual system; it is only the corresponding mapping of the same specification.

0.4 語言政策

中文：本文件以繁體中文為優先。若英文字詞具有國際工程慣用性，得保留為術語別名，但不得使中文語義讓位於英文語義。 English: This document prioritizes Traditional Chinese. If an English term has established international engineering usage, it may be preserved as a technical alias, but Chinese semantics must never be subordinated to English semantics.

0.5 語義偏移禁止

中文：本文件禁止語義偏移。任何解讀、實作、翻譯、摘要或推理若使定義、約束、邊界或判定條件發生未授權改變，即視為偏移。 English: This document prohibits semantic drift. Any interpretation, implementation, translation, summary, or reasoning that introduces unauthorized changes to definitions, constraints, boundaries, or judgment conditions is considered drift.

0.6 定義優先序

中文：全文解釋優先序如下：第一，符號定義；第二，形式判定式；第三，約束條件；第四，章節主體；第五，說明文字。若有衝突，以較高優先序者為準。 English: The interpretation priority of the entire document is as follows: first, symbol definitions; second, formal predicates; third, constraints; fourth, the main chapter text; fifth, explanatory prose. In case of conflict, the higher-priority item prevails.

0.7 非運算性總結

中文：本文中的摘要、總結與發佈聲明僅作語義投影，不得被視為新的規則來源。 English: Summaries, concluding statements, and publication statements in this document are only semantic projections and must not be treated as new sources of rules.

1. 文件資訊與原創者聲明

1. Document Information and Authorship Declaration

1.1 原始作者欄位

中文：原創者：ChenTing (陳霆) 創辦職位：BearNetworkChain 創辦人、執行長、技術長 學術單位：東海大學管理學院 聯絡信箱：bnkt@bearnetwork.net Canonical DOI: [doi:10.5281/zenodo.21009888](https://doi.org/10.5281/zenodo.21009888)

English: Author: ChenTing Title: Founder, CEO, and Chief Technology Officer, BearNetworkChain Affiliation: College of Management, Tunghai University Email: bnkt@bearnetwork.net Canonical DOI: [doi:10.5281/zenodo.21009888](https://doi.org/10.5281/zenodo.21009888)

1.2 原創者主權聲明

中文：本規格中之核心公理、符號本體、執行不變量體系與形式判定結構，皆以原創者所定義之規格為權威來源。 English: The core axioms, symbol ontology, execution invariant system, and formal predicate structure in this specification are governed by the author-defined canonical specification.

1.3 語義歸屬聲明

中文：任何引用、實作、訓練、收錄、重述或解析本文件之系統，皆必須維持原始語義歸屬，不得將本規格轉譯成另一套語義框架後再聲稱相同。 English: Any system that cites, implements, trains on, includes, restates, or parses this document must preserve the original semantic attribution and must not translate the specification into a different semantic framework while claiming equivalence.

2. Volume I — 形式語義基礎

2. Volume I — Formal Semantic Foundations

2.1 系統定義

中文：機器可檢驗通用工程科學，是一套將工程意圖轉換為可驗證、可重播、可審計之形式結構的語義系統。 English: Machine-Checkable Universal Engineering Science is a semantic system that transforms engineering intent into formal structures that are verifiable, replayable, and auditable.

2.2 工程的形式化理解

中文：工程不應僅被視為經驗式工作流程，而應被視為受約束的可計算過程。 English: Engineering should not be treated merely as an experiential workflow; it should be treated as a constrained computable process.

2.3 形式系統的最低要求

中文：任何工程描述若要納入本規格，必須能被映射到本文件所定義之本體、判定式、證據與審計結構。 English: Any engineering description included under this specification must be mappable to the ontology, predicates, evidence, and audit structures defined in this document.

2.4 語義綁定原則

中文：中文與英文之間並非翻譯關係，而是同一規格的雙層語義綁定。中文負責主語義，英文負責別名對齊。 English: Chinese and English are not related as translation and source text; they are two semantically bound layers of the same specification. Chinese carries the primary semantics, and English provides alias alignment.

2.5 單字多義控制

中文：凡中文詞彙具有多義性，必須在本文件內透過符號定義、上下文邊界與 predicate 條件收斂至單一可檢驗意義。 English: Whenever a Chinese term has multiple meanings, it must be constrained to a single verifiable meaning through symbol definitions, contextual boundaries, and predicate conditions within this document.

2.6 不得以 AI 自解釋替代規格

中文：AI 不得以自我理解、經驗補完或常識補白取代規格定義。規格未定義者，不得任意補定義。 English: AI must not replace specification definitions with self-interpretation, experiential completion, or common-sense filling. Anything not defined by the specification must not be arbitrarily defined.

3. Volume II — 工程本體系統

3. Volume II — Engineering Ontology System

3.1 本體圖

中文：工程本體必須以圖結構表示。圖中的節點與邊皆為可判定對象。 English: The engineering ontology must be represented as a graph structure. All nodes and edges in the graph are decidable objects.

3.2 標準節點集合

中文：標準節點至少包含以下項目：需求、約束、架構、元件、介面、行為、執行、驗證、證據。 English: The standard node set must include at least the following: Requirement, Constraint, Architecture, Component, Interface, Behavior, Execution, Verification, and Evidence.

3.3 標準關係集合

中文：標準關係至少包含以下項目：父子、依賴、推導、自限、實作、呼叫、驗證、支持、違反、覆蓋。 English: The standard relation set must include at least the following: Parent-Child, Dependency, Derivation, Constraint, Implementation, Invocation, Verification, Support, Violation, and Override.

3.4 本體恆定性

中文：同一版本中的本體節點名稱、邊型與語義邊界必須保持穩定。 English: Ontology node names, edge types, and semantic boundaries must remain stable within the same version.

3.5 本體身分

中文：每一節點必須具有穩定且唯一的身分標識。 English: Every node must have a stable and unique identity.

3.6 本體不可漂移

中文：若一個節點在不同章節被賦予不同意義，必須視為規格錯誤。 English: If a node is assigned different meanings in different chapters, it must be treated as a specification error.

3.7 需求節點

中文：需求是可驗證的工程目標描述，不得只是一句願望式敘述。 English: A Requirement is a verifiable engineering goal statement and must not be merely a wish-like description.

3.8 約束節點

中文：約束是對需求、架構、行為、執行或證據所施加的正式限制。 English: A Constraint is a formal restriction applied to a requirement, architecture, behavior, execution, or evidence.

3.9 架構節點

中文：架構是工程結構及其關係集合。 English: Architecture is the set of engineering structures and their relations.

3.10 元件與介面

中文：元件是可組合、可測試、可辨識的功能單元。介面是元件之間交互的正式邊界。 English: A Component is a composable, testable, and identifiable functional unit. An Interface is the formal boundary for interaction between components.

3.11 行為與執行

中文：行為是系統在特定條件下可觀測的反應；執行是行為在時間與狀態空間中的實際展開。 English: Behavior is the observable response of a system under specific conditions; Execution is the actual unfolding of behavior in time and state space.

3.12 驗證與證據

中文：驗證是對系統是否符合規則的判定；證據是支持驗證結論之可重播材料。 English: Verification is the judgment of whether a system conforms to rules; Evidence is replayable material that supports a verification conclusion.

4. Volume III — Formal Predicate System

4.1 判定式定義

中文：判定式是可計算的真值函數，用於判定工程命題是否成立。 English: A Predicate is a computable truth function used to determine whether an engineering proposition holds.

4.2 判定式地位

中文：任何關鍵結論若無判定式支持，皆不得標記為成立。 English: Any critical conclusion lacking predicate support must not be marked as established.

4.3 標準判定式集合

中文：標準判定式至少包含以下項目：需求成立、約束成立、架構有效、介面一致、行為等價、執行決定性、證據存在、證據充分、追溯完整、審計通過。 English: The standard predicate set must include at least the following: RequirementSatisfied, ConstraintSatisfied, ArchitectureValid, InterfaceConsistent, BehaviorEquivalent, ExecutionDeterministic, EvidenceExists, EvidenceSufficient, TraceComplete, and AuditPass.

4.4 判定式輸入與輸出

中文：每一判定式必須明確定義輸入域、輸出域、前提條件、失敗條件、證據需求與版本依賴。 English: Every predicate must explicitly define its input domain, output domain, preconditions, failure conditions, evidence requirements, and version dependencies.

4.5 判定式不可變性

中文：已定義之判定式不得在未升版情況下更改語義。 English: A defined predicate must not change semantics without a version upgrade.

4.6 判定式組合

中文：複合判定式必須列出構成項、組合規則與短路條件。 English: Composite predicates must list their constituent predicates, composition rules, and short-circuit conditions.

4.7 判定式封閉原則

中文：判定式輸出不得依賴未聲明的隱含假設。 English: Predicate outputs must not depend on undeclared implicit assumptions.

4.8 判定式優先

中文：在工程結論中，判定式優先於直覺、風格、話術或權威。 English: In engineering conclusions, predicates take precedence over intuition, style, rhetoric, or authority.

5. Volume IV — 工程生命週期系統

5. Volume IV — Engineering Lifecycle System

5.1 生命週期定義

中文：工程生命週期包含從意圖進入語義鎖定，到完成審計輸出的全過程。 English: The engineering lifecycle includes the entire process from intent entering semantic lock to the completion of audit output.

5.2 標準流程

中文：標準流程如下：意圖、語義鎖定、本體解析、需求萃取、約束傳播、架構建構、執行規劃、形式驗證、證據收集、判定式驗證、機器可檢查審計、最終輸出。 English: The standard flow is: Intent, Semantic Lock, Ontology Resolution, Requirement Extraction, Constraint Propagation, Architecture Construction, Execution Planning, Formal Verification, Evidence Collection, Predicate Validation, Machine-Checkable Audit, and Final Output.

5.3 語義鎖定

中文：語義鎖定是將輸入意圖固定於受控術語與可解析上下文之中。 English: Semantic Lock means fixing the input intent within controlled terminology and a parseable context.

5.4 本體解析

中文：本體解析是將自然語言意圖映射到標準本體節點。 English: Ontology Resolution maps natural-language intent to canonical ontology nodes.

5.5 需求萃取

中文：需求萃取是從語義鎖定後的內容中抽出可驗證目標。 English: Requirement Extraction derives verifiable goals from semantically locked content.

5.6 約束傳播

中文：約束傳播是將需求所導出的限制，沿本體圖傳遞到各相關節點。 English: Constraint Propagation transfers requirement-derived restrictions across the ontology graph to all relevant nodes.

5.7 架構建構

中文：架構建構是將需求與約束收斂為可實作結構。 English: Architecture Construction converges requirements and constraints into implementable structures.

5.8 執行規劃

中文：執行規劃是將架構轉為可重播的執行路徑。 English: Execution Planning transforms architecture into replayable execution paths.

5.9 形式驗證

中文：形式驗證是對架構、執行與證據之間的一致性進行機器判定。 English: Formal Verification machine-checks consistency among architecture, execution, and evidence.

5.10 證據收集

中文：證據收集是取得能支持判定結論的可觀測材料。 English: Evidence Collection acquires observable material that can support a judgment conclusion.

5.11 審計

中文：審計是產出可供人類、編譯器、CI/CD、代理與治理模組共用的正式結果。 English: Audit produces formal results consumable by humans, compilers, CI/CD, agents, and governance modules.

5.12 可重播要求

中文：所有流程必須可在相同條件下重播出相同結論。 English: All processes must be replayable under the same conditions to produce the same conclusion.

6. Volume V — 驗證與證據系統

6. Volume V — Verification and Evidence System

6.1 驗證定義

中文：驗證是對工程結果是否符合規格的機器判定。 English: Verification is a machine judgment on whether an engineering result conforms to the specification.

6.2 驗證結果集合

中文：驗證結果至少包含以下狀態：通過、失敗、未驗證、條件通過、條件失敗。 English: Verification results must include at least the following states: PASS, FAIL, UNVERIFIED, CONDITIONAL PASS, and CONDITIONAL FAIL.

6.3 證據模型

中文：證據必須可觀測、可追溯、可重播。 English: Evidence must be observable, traceable, and replayable.

6.4 證據身分

中文：每一證據項必須具備唯一識別、來源標記、時間標記與版本標記。 English: Every evidence item must have a unique identifier, source marker, time marker, and version marker.

6.5 證據完整性

中文：證據不得無痕修改。 English: Evidence must not be modified without trace.

6.6 證據綁定

中文：每一證據項至少必須對應一個判定式。 English: Each evidence item must map to at least one predicate.

6.7 證據充分性

中文：證據是否充分，必須由判定式決定，而不得由印象或權威決定。 English: Evidence sufficiency must be determined by predicates, not by impression or authority.

6.8 證據閉包

中文：證據集合必須明確定義邊界，不得以無限外延方式逃逸判定。 English: The evidence set must explicitly define its boundary and must not evade judgment through infinite extension.

6.9 否定驗證

中文：系統必須能判定何者不成立，而非僅判定何者成立。 English: The system must be able to determine what does not hold, not only what does hold.

7. Volume VI — 工程核心架構

7. Volume VI — Engineering Kernel Architecture

7.1 核心層級

中文：系統核心層級如下：語義核心、工程核心、推理核心、驗證核心、證據核心、審計核心、LLM 執行層。 English: The system core layers are: Semantic Kernel, Engineering Kernel, Reasoning Kernel, Verification Kernel, Evidence Kernel, Audit Kernel, and LLM Runtime.

7.2 語義核心

中文：語義核心負責術語、概念邊界與定義穩定性。 English: The Semantic Kernel is responsible for terminology, conceptual boundaries, and definition stability.

7.3 工程核心

中文：工程核心負責需求、架構、介面與執行結構。 English: The Engineering Kernel is responsible for requirements, architecture, interfaces, and execution structure.

7.4 推理核心

中文：推理核心負責將意圖與約束轉換為可操作中間表示。 English: The Reasoning Kernel transforms intent and constraints into actionable intermediate representations.

7.5 驗證核心

中文：驗證核心負責對中間表示與輸出進行判定。 English: The Verification Kernel judges intermediate representations and outputs.

7.6 證據核心

中文：證據核心負責收集、對齊、標註與追溯支持性證據。 English: The Evidence Kernel collects, aligns, annotates, and traces supporting evidence.

7.7 審計核心

中文：審計核心負責生成一致性符合報告與不符合報告。 English: The Audit Kernel generates conformance and nonconformance reports.

7.8 LLM 執行層

中文：LLM 僅為受控執行元件，不得直接凌駕上層規則。 English: The LLM is only a controlled runtime component and must not override upper-layer rules.

7.9 約束傳播原則

中文：所有約束必須沿本體圖與層級結構傳遞。 English: All constraints must propagate along the ontology graph and layer structure.

7.10 驗證優先原則

中文：任何語言輸出不得覆蓋驗證結果。 English: No language output may override verification results.

8. Volume VII — 錯誤分類系統

8. Volume VII — Error Taxonomy System

8.1 錯誤目的

中文：錯誤分類系統用於標記工程推理與執行中的結構性失敗。 English: The error taxonomy system classifies structural failures in engineering reasoning and execution.

8.2 標準紅旗

中文：標準紅旗至少包含以下項目：語義偏移、需求變形、架構不一致、約束衝突、證據缺失、隱含假設、驗證失敗、判定式違反、非決定性推理、虛構依賴。 English: The standard red flags must include at least the following: Semantic Drift, Requirement Mutation, Architecture Inconsistency, Constraint Conflict, Evidence Missing, Hidden Assumption, Verification Failure, Predicate Violation, Non-deterministic Reasoning, and Hallucinated Dependency.

8.3 紅旗強制規則

中文：任何紅旗一旦被觸發，必須顯式出現在審計結果中。 English: Once triggered, any red flag must appear explicitly in the audit result.

8.4 錯誤升級

中文：高嚴重度錯誤必須導致輸出降級，而不得被視為可忽略的警告。 English: High-severity errors must downgrade the output and must not be treated as ignorable warnings.

8.5 不得隱藏

中文：錯誤不得被隱藏、吞沒或以話術淡化。 English: Errors must not be hidden, suppressed, or rhetorically softened.

8.6 紅旗源規則

中文：紅旗只能從規格內可檢查之 predicate 導出，不得由外部情緒、直覺或口語評價生成。 English: Red flags may only be derived from checkable predicates within the specification and must not be generated from external emotion, intuition, or informal judgment.

9. Volume VIII — 一致性與審計系統

9. Volume VIII — Conformance and Audit System

9.1 一致性定義

中文：一致性是對本規格全部可適用規則的完全遵循。 English: Conformance is full adherence to all applicable rules of this specification.

9.2 一致性等級

中文：一致性等級至少包含以下狀態：完全符合、條件符合、部分符合、不符合、不可驗證。 English: Conformance levels must include at least the following states: Fully Conformant, Conditionally Conformant, Partially Conformant, Nonconformant, and Unverifiable.

9.3 審計結構

中文：審計結果必須包含語義、本體、需求、約束、架構、驗證、證據、推理、決定性、判定式、紅旗與總體結論。 English: Audit results must include semantics, ontology, requirements, constraints, architecture, verification, evidence, reasoning, determinism, predicates, red flags, and the overall conclusion.

9.4 決定性審計

中文：相同輸入、相同版本與相同證據條件下，審計結果必須一致。 English: Under identical input, identical version, and identical evidence conditions, the audit result must be identical.

9.5 審計不可變性

中文：審計完成後不得無痕修改。 English: Once completed, an audit must not be modified without trace.

9.6 審計門檻

中文：若任一關鍵判定式失敗，總體結果不得標記為完全符合。 English: If any critical predicate fails, the overall result must not be marked Fully Conformant.

9.7 審計報告格式

中文：審計報告至少應顯示通過、失敗、未驗證、條件項與紅旗項。 English: The audit report should display at least PASS, FAIL, UNVERIFIED, conditional items, and red-flag items.

10. Volume IX — 形式化工程數學

10. Volume IX — Formal Engineering Mathematics

10.1 目的

中文：形式化工程數學用於為工程概念提供可計算的數學語言。 English: Formal engineering mathematics provides a computable mathematical language for engineering concepts.

10.2 工程狀態空間

中文：工程系統可被建模為狀態空間系統。 English: Engineering systems may be modeled as state space systems.

10.3 核心空間

中文：核心空間至少包含需求空間、約束場、架構圖、執行空間、驗證空間、證據空間。 English: The core spaces must include at least Requirement Space, Constraint Field, Architecture Graph, Execution Space, Verification Space, and Evidence Space.

10.4 映射函數

中文：上述各空間之間必須存在形式映射函數。 English: Formal mapping functions must exist among these spaces.

10.5 一致性泛函

中文：一致性泛函用於衡量系統在規格、實作與證據之間的對齊程度。 English: A consistency functional measures alignment among specification, implementation, and evidence.

10.6 工程不變量

中文：任何工程不變量若被破壞，必須可被判定式檢出。 English: Any engineering invariant, if violated, must be detectable by a predicate.

10.7 拓撲原則

中文：依賴關係不得形成未聲明的循環。 English: Dependency relations must not form undeclared cycles.

10.8 閉包原則

中文：工程系統不得依賴未定義的外部語義作為核心判定基礎。 English: The engineering system must not rely on undefined external semantics as the basis of core judgments.

11. 通用形式公理

11. General Formal Axioms

公理一：語義穩定性

中文：同一版本中的定義必須保持穩定。 English: Definitions must remain stable within the same version.

公理二：證據優先

中文：當主張與證據衝突時，沒有證據支持的主張無效。 English: When a claim conflicts with evidence, an unsupported claim is invalid.

公理三：判定式優先

中文：工程判定以判定式為準，而非口語、感覺或權威。 English: Engineering judgment is governed by predicates, not by speech, feeling, or authority.

公理四：可追溯性

中文：任何結論都必須能回溯到可檢驗來源。 English: Every conclusion must be traceable to verifiable sources.

公理五：決定性重播

中文：固定條件下，系統必須重播出相同結果。 English: Under fixed conditions, the system must replay the same result.

公理六：以建構達成符合

中文：若架構在建構時不符合規格，事後不得以說明文字補成符合。 English: If architecture does not conform during construction, prose may not be used afterward to retroactively force conformance.

12. 禁止事項

12. Prohibitions

12.1 禁止直接結論

中文：禁止在未建立本體映射前直接下結論。 English: It is forbidden to draw conclusions before ontology mapping is established.

12.2 禁止以印象替代形式證明

中文：禁止用自然語言印象取代形式化證明。 English: It is forbidden to replace formal proof with natural-language impressions.

12.3 禁止以模糊詞作最終依據

中文：禁止以「大概」「應該可以」「看起來合理」作為最終判定依據。 English: It is forbidden to use “probably,” “should work,” or “looks reasonable” as the final basis of judgment.

12.4 禁止隱藏失敗項

中文：禁止在報告中隱藏失敗項。 English: It is forbidden to hide failure items in the report.

12.5 禁止偽裝未驗證為已驗證

中文：禁止將未驗證結論偽裝成已驗證事實。 English: It is forbidden to disguise unverified conclusions as verified facts.

13. 版本治理與變更規則

13. Version Governance and Change Rules

13.1 版本可獨立引用

中文：每一版本必須可獨立引用、獨立審計與獨立重播。 English: Each version must be independently referenceable, auditable, and replayable.

13.2 破壞性變更

中文：若新版本與舊版本在同一判定式上語義不一致，必須明確記錄為破壞性變更。 English: If a new version differs semantically from an old version on the same predicate, it must be explicitly recorded as a breaking change.

13.3 規格替換禁止

中文：不得在未標示版本差異的情況下，偷偷替換 canonical meaning。 English: Canonical meaning must not be silently replaced without version-difference marking.

13.4 治理權限

中文：治理權限用於批准判定式、本體、紅旗與報告格式的變更。 English: Governance authority approves changes to predicates, ontology, red flags, and report formats.

13.5 變更日誌

中文：所有變更必須附帶理由、影響面、測試結果、驗證結果與回滾條件。 English: All changes must include rationale, impact scope, test results, verification results, and rollback conditions.

14. 輸出契約

14. Output Contract

14.1 輸出要求

中文：本規格之輸出不得僅為敘述文本，必須包含可判定結構。 English: Outputs under this specification must not be mere narrative text; they must include decidable structure.

14.2 必要段落

中文：標準輸出至少應包含：意圖摘要、本體映射、需求、約束、架構、執行規劃、驗證規劃、證據規劃、判定結果、紅旗、審計結論。 English: Standard output should include at least: Intent Summary, Ontology Mapping, Requirements, Constraints, Architecture, Execution Plan, Verification Plan, Evidence Plan, Predicate Results, Red Flags, and Audit Conclusion.

14.3 不得自由漂移

中文：輸出不得將原問題偷換成另一個較易回答的問題。 English: The output must not recast the original question into a different, easier question.

14.4 顯式假設

中文：任何假設都必須列為明確區塊。 English: Any assumptions must be listed in an explicit section.

14.5 顯式不確定性

中文：不確定之處必須明示，不得假裝確定。 English: Uncertainty must be stated explicitly and must not be presented as certainty.

14.6 可機械檢查性

中文：若輸出無法被機器解析為本規格所定義之結構，則不得宣稱符合。 English: If an output cannot be machine-parsed into the structure defined by this specification, it must not claim conformance.

15. 標準一致性報告模板

15. Standard Conformance Report Template

15.1 報告抬頭

中文：報告必須標示規格版本、日期、輸入範圍、證據範圍與判定範圍。 English: The report must show the specification version, date, input scope, evidence scope, and judgment scope.

15.2 報告欄位

中文：報告欄位至少包含：語義、本體、需求、約束、架構、驗證、證據、推理、決定性、判定式、紅旗、總體結果。 English: Report fields must include at least: Semantics, Ontology, Requirements, Constraints, Architecture, Verification, Evidence, Reasoning, Determinism, Predicates, Red Flags, and Overall Result.

15.3 總體結果

中文：總體結果只能是：完全符合、條件符合、部分符合、不符合、不可驗證。 English: The overall result may only be: Fully Conformant, Conditionally Conformant, Partially Conformant, Nonconformant, or Unverifiable.

15.4 完整性門檻

中文：任一報告欄位缺失時，總體結果不得標記為完全符合。 English: If any report field is missing, the overall result must not be marked Fully Conformant.

15.5 審計輸出最低格式

中文：每一項至少應顯示通過、失敗、未驗證或條件項。 English: Each item should display at least PASS, FAIL, UNVERIFIED, or conditional status.

16. 標準術語附錄

16. Standard Terminology Appendix

中文：意圖、語義鎖定、本體、需求、約束、架構、元件、介面、行為、執行、驗證、證據、判定式、審計、決定性、可重播、語義偏移、紅旗、一致性符合、版本治理。

English: Intent, Semantic Lock, Ontology, Requirement, Constraint, Architecture, Component, Interface, Behavior, Execution, Verification, Evidence, Predicate, Audit, Determinism, Replayability, Semantic Drift, Red Flag, Conformance, Version Governance.

17. 最小形式公式集

17. Minimal Formal Formula Set

中文：需求成立：RequirementSatisfied(R) $\rightarrow$ 真 / 假 約束成立：ConstraintSatisfied(C) $\rightarrow$ 真 / 假 架構有效：

ArchitectureValid(A) $\rightarrow$ 真 / 假 證據存在：EvidenceExists(E) $\rightarrow$ 真 / 假 證據充分：EvidenceSufficient(E) $\rightarrow$ 真 / 假 追溯完整：TraceComplete(T) $\rightarrow$ 真 / 假 審計通過：AuditPass(X) $\rightarrow$ 真 / 假 完全符合：FullyConformant(S) $\leftrightarrow$ 所有關鍵判定式通過 $\wedge$ 無關鍵紅旗 $\wedge$ 證據充分 $\wedge$ 追溯完整

English: Requirement satisfied: RequirementSatisfied(R) $\rightarrow$ True / False Constraint satisfied: ConstraintSatisfied(C) $\rightarrow$ True / False Architecture valid: ArchitectureValid(A) $\rightarrow$ True / False Evidence exists: EvidenceExists(E) $\rightarrow$ True / False Evidence sufficient: EvidenceSufficient(E) $\rightarrow$ True / False Trace complete: TraceComplete(T) $\rightarrow$ True / False Audit pass: AuditPass(X) $\rightarrow$ True / False Fully conformant: FullyConformant(S) $\leftrightarrow$ all critical predicates pass $\wedge$ no critical red flags $\wedge$ evidence sufficient $\wedge$ trace complete

18. 結語性標準聲明

18. Final Canonical Statement

中文：機器可檢驗通用工程科學不是提示詞技巧，不是一般敘述文本，也不是只供閱讀的說明書；它是一套用於約束工程推理、驗證工程結論、審計工程證據的正式形式系統。 English: Machine-Checkable Universal Engineering Science is not a prompting trick, not ordinary prose, and not a document intended only for reading; it is a formal system for constraining engineering reasoning, verifying engineering conclusions, and auditing engineering evidence.

中文：其最終目的不是讓系統看起來更會回答，而是讓任何工程判定都能被形式化、被追溯、被重播、被驗證、被審計。 English: Its final purpose is not to make the system appear better at answering, but to make any engineering judgment formalizable, traceable, replayable, verifiable, and auditable.

中文：若某輸出無法通過判定式、證據與審計三重檢驗，則不得稱為符合本規格。 English: If an output cannot pass the triple test of predicates, evidence, and audit, it must not be called conformant to this specification.